Supplementary Material

Anonymous Author(s)

1 Theoretical Justification of UCB-Maximizing Prior Aggregation

In traditional reinforcement learning and bandit problems, the Upper Confidence Bound (UCB) strategy selects a discrete optimal state by maximizing a scalar acquisition function. However, in the proposed framework, the shared MLP_θ maps regional statistics (mean and variance) of k randomly sampled anchors in $\mathcal{Z}_u = \{z_1, \ldots, z_k\}$ into a d_m-dimensional continuous embedding space.

In this section, we provide a formal mathematical proof demonstrating that applying a dimension-wise Max-Pooling operation across these latent representations computes the *supremum* (least upper bound) of the exploration signals. We further prove that this operation consistently dominates any discrete single-anchor selection, thereby perfectly operationalizing the exploration maximization goal of the UCB policy in a high-dimensional continuous space.

1.1 Definitions and Theorem

Assumption 1 (Monotonicity in Latent Exploration). We assume the parameterized mapping MLP_θ implicitly learns a monotonic transformation with respect to the regional epistemic uncertainty (i.e., variance L_{var}^{u,z_n}). Therefore, a larger value in the latent dimension d corresponds to a higher upper confidence bound for that specific semantic factor.

Definition 1 (Partial Ordering in Latent Space). Let $V = \{\mathbf{v}^{(1)}, \mathbf{v}^{(2)}, \ldots, \mathbf{v}^{(k)}\}$ be the set of exploration prior vectors generated for the k sampled anchors, where each vector is defined as $\mathbf{v}^{(n)} = \text{MLP}_\theta\left(\left[L_{\text{mean}}^{u,z_n} \,;\, L_{\text{var}}^{u,z_n}\right]\right) \in \mathbb{R}^{d_m}$. We define a partial order $\preceq$ on $\mathbb{R}^{d_m}$ such that for any two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{d_m}$:

$$\mathbf{x} \preceq \mathbf{y} \iff x_d \leq y_d \quad \forall d \in \{1, \ldots, d_m\}. \tag{1}$$

Following the standard representation paradigm in deep pooling networks [1], we assume the parameterized space implicitly orthogonalizes the representations, such that each dimension d roughly corresponds to a decoupled semantic factor in the latent preference space.

Theorem 1 (Supremum of Exploration Signals). The dimension-wise Max-Pooling operation, defined as $\mathbf{L}^u = \max_{n=1}^{k} \mathbf{v}^{(n)}$, computes the exact supremum (least upper bound) of the exploration set V.

1.2 Formal Proof

Proof. The proof consists of two parts: first demonstrating that $\mathbf{L}^u$ is a valid upper bound, and second demonstrating that it is the *least* upper bound.

Part I: $\mathbf{L}^u$ is an upper bound of V. By the definition of the dimension-wise Max-Pooling operation, the d-th dimension of the output vector $\mathbf{L}^u$ is computed as:

$$L_d^u = \max_{1 \leq n \leq k} v_d^{(n)}. \tag{2}$$

It trivially follows that for any specific anchor index $n \in \{1, \ldots, k\}$ and any dimension $d \in \{1, \ldots, d_m\}$, the relation $v_d^{(n)} \leq L_d^u$ holds. According to Definition 1, this implies $\mathbf{v}^{(n)} \preceq \mathbf{L}^u$ for all $\mathbf{v}^{(n)} \in V$. Therefore, $\mathbf{L}^u$ is a valid upper bound for the set V.

Part II: $\mathbf{L}^u$ is the least upper bound. Suppose there exists another arbitrary upper bound $\mathbf{b} \in \mathbb{R}^{d_m}$ such that $\mathbf{v}^{(n)} \preceq \mathbf{b}$ for all $n \in \{1, \ldots, k\}$. By Definition 1, this implies:

$$v_d^{(n)} \leq b_d \quad \forall n \in \{1, \ldots, k\},\ \forall d \in \{1, \ldots, d_m\}. \tag{3}$$

Since b_d bounds every individual element in the set $\{v_d^{(1)}, \ldots, v_d^{(k)}\}$, it must inherently be greater than or equal to the maximum value of this set. Thus, we have:

$$\max_{1 \leq n \leq k} v_d^{(n)} \leq b_d \implies L_d^u \leq b_d \quad \forall d \in \{1, \ldots, d_m\}. \tag{4}$$

Applying Definition 1 again, this yields $\mathbf{L}^u \preceq \mathbf{b}$.

Since $\mathbf{L}^u$ is a valid upper bound and is mathematically less than or equal to any other conceivable upper bound $\mathbf{b}$, it is strictly the supremum of V, denoted as $\mathbf{L}^u = \sup(V)$. ■

Remark: While the algebraic derivation of the supremum follows directly from the definition of Max-Pooling, its theoretical significance lies in mapping the discrete topological space of traditional bandit algorithms into a continuous representation. The supremum vector $\mathbf{L}^u$ acts as a "virtual super-anchor" that analytically envelops the epistemic uncertainty boundaries of all individual anchors simultaneously, enabling a one-shot exploration constraint during the generative process.

1.3 Implications for Generative Conditioning

Corollary 1 (Maximized Exploration Efficacy). During the subsequent conditional diffusion process, the composite prior $\mathbf{L}^u$ guides the generation via neural projections (e.g., Cross-Attention mechanisms). Assuming a standard non-negative weight projection (e.g., post-Softmax attention probabilities) represented by a weight vector $\mathbf{w} \in \mathbb{R}_{\geq 0}^{d_m}$, the generative conditioning signal (inner product) satisfies:

$$\mathbf{w}^\top \mathbf{v}^{(n)} \leq \mathbf{w}^\top \mathbf{L}^u, \quad \forall n \in \{1, \ldots, k\}. \tag{5}$$

Remark: This corollary formally demonstrates that within a non-negative feature subspace, the composite upper-bound vector $\mathbf{L}^u$ guarantees a generative conditioning signal that theoretically upper-bounds the signal provided by any single discretely selected anchor. Therefore, the dimension-wise Max-Pooling operation successfully operationalizes the UCB maximization objective, actively steering the diffusion process toward the absolute boundaries of epistemic uncertainty.

2 Datasets

To comprehensively evaluate the effectiveness and generalizability of our proposed framework, we conduct experiments on three widely adopted real-world benchmark datasets: *Amazon-Baby*, *Amazon-Sports*, and *TikTok*. These datasets are intentionally selected to encompass diverse application scenarios, ranging from traditional e-commerce retail to modern micro-video sharing platforms.

Specifically, *Amazon-Baby* and *Amazon-Sports* are subsets of the prominent Amazon review dataset, containing user-item interaction records alongside visual and textual modalities (i.e., product images and descriptions). *TikTok* introduces a more complex micro-video recommendation scenario, encompassing visual, textual, and additionally acoustic modalities. Crucially, all three datasets exhibit extreme interaction sparsity (over 99.88%), providing an ideal testbed to verify our model's capability in resolving epistemic sparsity via diffusion-based prior aggregation.

To ensure strictly fair baseline comparisons and avoid preprocessing biases, we directly utilize the publicly available, pre-extracted multimodal features provided by established benchmark studies [2, 3]. According to their original feature extraction protocols, the visual features of product images and video frames were extracted using the pre-trained ResNet [4] architecture. For the textual modality, semantic embeddings of product descriptions and video captions were encoded via the pre-trained BERT [5] model. Furthermore, to capture the auditory signals in the *TikTok* dataset, acoustic features were extracted utilizing the pre-trained VGGish [6] network. We adopt rigorous data partitioning schemes (e.g., 8:1:1 split for training, validation, and testing) [7–9] consistent with these prior works.

Dataset	Baby		Sports		TikTok		
Modality	V	T	V	T	V	T	A
Embed Dim	4096	384	4096	384	128	768	128
User	19,445		35,598		9,308		
Item	7,050		18,357		4,710		
Interactions	160,792		296,337		68,722		
Sparsity	99.88%		99.95%		99.89%		

Table 1: Statistics of three real-world datasets with Visual(V), Textual(T), and Acoustic(A) contents of items.

References

[1] Charles R Qi, Hao Su, Kaichun Mo, and Leonidas J Guibas. Pointnet: Deep learning on point sets for 3d classification and segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 652–660, 2017.

[2] Wei Wei, Chao Huang, Lianghao Xia, and Chuxu Zhang. Multi-modal self-supervised learning for recommendation. In *Proceedings of the ACM web conference 2023*, pages 790–800, 2023.

[3] Yangqin Jiang, Lianghao Xia, Wei Wei, Da Luo, Kangyi Lin, and Chao Huang. DiffMM: Multi-modal diffusion model for recommendation. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pages 7591–7599, 2024.

[4] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.

[5] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pages 4171–4186, 2019.

[6] Shawn Hershey, Sourish Chaudhuri, Daniel PW Ellis, Jort F Gemmeke, Aren Jansen, R Channing Moore, Manoj Plakal, Devin Platt, Rif A Saurous, Bryan Seybold, et al. Cnn architectures for large-scale audio classification. In *2017 ieee international conference on acoustics, speech and signal processing (icassp)*, pages 131–135. IEEE, 2017.

[7] Xin Zhou, Hongyu Zhou, Yong Liu, Zhiwei Zeng, Chunyan Miao, Pengwei Wang, Yuan You, and Feijun Jiang. Bootstrap latent representations for multi-modal recommendation. In *Proceedings of the ACM web conference 2023*, pages 845–854, 2023.

[8] Xin Zhou and Zhiqi Shen. A tale of two graphs: Freezing and denoising graph structures for multimodal recommendation. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages 935–943, 2023.

[9] Guipeng Xv, Xinyu Li, Ruobing Xie, Chen Lin, Chong Liu, Feng Xia, Zhanhui Kang, and Leyu Lin. Improving multi-modal recommender systems by denoising and aligning multi-modal content and user feedback. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 3645–3656, 2024.